MAE 4730
10/5/18
Lecture 18

Today:
① odds + ends
② shaken inverted pendulum

How many independent equations of mechanics from one FBD?

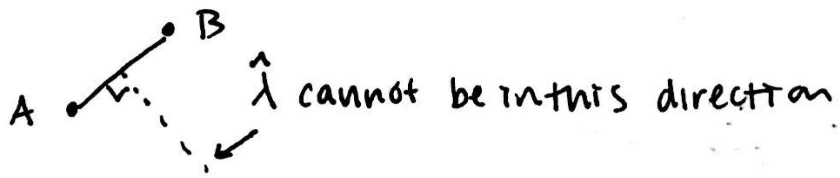
in 2D

- ANS: 3
- ex) $\{LMB\} \cdot \hat{\imath}$
  $\{LMB\} \cdot \hat{\jmath}$
  $\{AMB_{/0}\} \cdot \hat{k}$

  $LMB \Leftrightarrow \sum \vec{F}^{ext} = m_{tot} \vec{a}_G$
  "Force in x direction" ....

  $AMB_{/0} \Leftrightarrow \sum \vec{M}^{ext}_{/0} = \sum \vec{r}_{i/0} \times m_i \vec{a}_{i/\mathcal{F}}$
  "Moments about 0"

- or) $LMB \cdot \hat{\imath}$
  $LMB \cdot \hat{\jmath}$
  $AMB_{/G}$
- or) $\{LMB\} \cdot \vec{a}$
  $\{LMB\} \cdot \vec{b}$ where $\vec{a} \times \vec{b} \neq 0$ (not parallel)
  $AMB_{/C}$ C is any point
- or) $\{AMB\}_{/A}$ A, B, C non colinear – otherwise arbitrary
  $\{AMB\}_{/B}$
  $\{AMB\}_{/C}$ $\Rightarrow LMB$!
- or) $AMB_{/A}$ A & B any 2 different pts.
  $AMB_{/B}$
  $LMB \cdot \hat{\lambda}$



$\hat{\lambda}$ not $\parallel$ to $\hat{k} \times \vec{r}_{AB}$

$\hat{\lambda} \times (\hat{k} \times \vec{r}_{AB}) \neq 0$

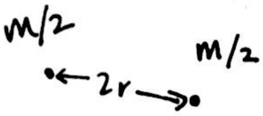
Moment of Inertia (2D)

$$I^G \equiv \begin{bmatrix} \sum m_i (\vec{r}_{i/G})^2 \\ \int d_{/G}^2 \, dm \end{bmatrix}$$

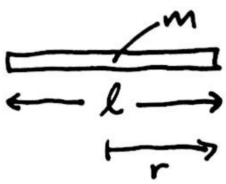
examples

• pt mass $I^G = 0$

hoop $I^G = r^2 m$

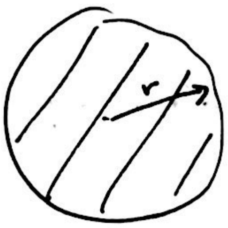


2 pt mass $I^G = mr^2$



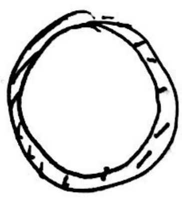
uniform stick $I^G = \frac{ml^2}{12} = \frac{mr^2}{3}$

Active Learning:



uniform disk
radius r
mass m

strips $\longrightarrow$

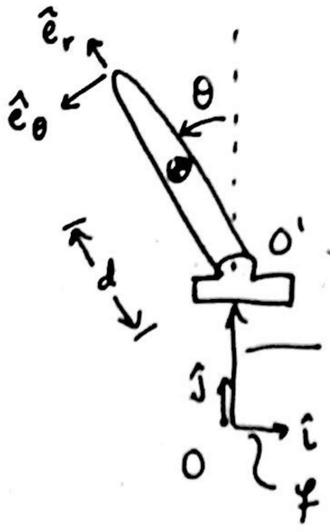


Area density $\rho = \frac{m}{\text{area}} = \frac{m}{\pi r^2}$

$I^G = \int r'^2 dm$

$$I^G = \int_0^r r'^2 \underbrace{\underbrace{\left(\frac{m}{\pi r^2}\right)}_{\rho} \underbrace{dA}_{2\pi r' dr'}}_{dm}$$

$I^G = \frac{1}{2} mr^2$



goes up + down powered by motor

given

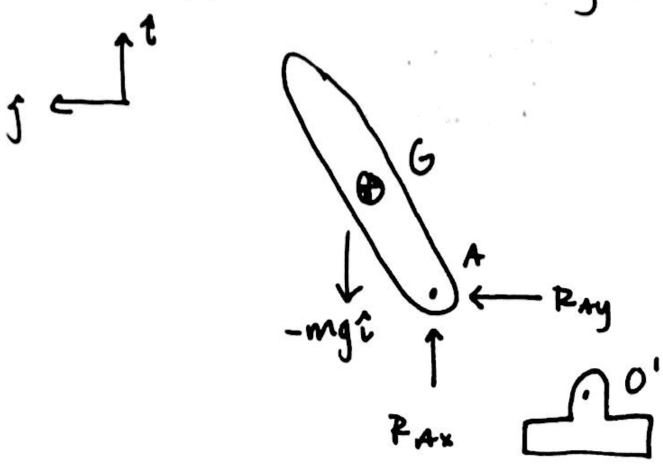
$y_{O'/O} = h\sin(\omega_m t) \neq \dot\theta$

angular velocity of motor

$-\omega_m^2 h\sin(\omega_m t)$

Last class we (all but) found $\boxed{\ddot\theta = \frac{(+g + a_{O'/O})\,d\,m}{I^G + md^2}\sin\theta}$

DAE Approach to finding EoM



LMB:

$$\sum \vec F^{ext} = m_{tot}\vec a_G$$

$$-mg\hat\imath + R_x\hat\imath + R_y\hat\jmath = m_{tot}(\ddot x\hat\imath + \ddot y\hat\jmath)$$

AMB/G

$$\sum M^{ext}_{/G} = I^G\ddot\theta\hat k$$

$$\underbrace{\vec r_{A/G}}_{-d\hat e_r}\times(R_{Ax}\hat\imath + R_{Ay}\hat\jmath)$$

Maximal coordinates: x_G, y_G, θ

3 equations linear in $\ddot x_G, \ddot y_G, \ddot\theta_G$

also linear in R_{Ax}, R_{Ay}

Const.

$$\vec r_A = \vec r_{O'} \Rightarrow \vec r_{A/O'} = \vec 0$$

$$\frac{d^2\vec r_{A/O}}{dt^2} = \vec 0$$

$$\boxed{\vec a_A = \vec a_{O'}}$$

Constraint eq: $\vec{a}_A = \vec{a}_0$ (2 scalar)

$\vec{a}_0$ given: $-\omega_m^2 h \sin(\omega_m t)\hat{\imath}$

$$\vec{a}_A = \vec{a}_G + \vec{a}_{A/G} = \underbrace{(\ddot{x}_G\hat{\imath} + \ddot{y}_G\hat{\jmath})}_{\vec{a}_G} + (-\dot{\theta}^2\vec{r}_{A/G} + \ddot{\theta}\hat{k}\times\vec{r}_{A/G})$$

$$\vec{r}_{A/G} = -d\underbrace{\hat{e}_r}_{\cos\theta\hat{\imath}+\sin\theta\hat{\jmath}}$$

$\Rightarrow$ 2 scalar eqs in $\ddot{x}_G, \ddot{y}_G, \ddot{\theta}$

reaction force contribution

$$\Rightarrow \begin{array}{c} \\ 3 \\ \\ \end{array}\left[\begin{array}{ccc|cc} m & & & \cdot & \cdot \\ & m & & \cdot & \cdot \\ & & I^G & \cdot & \cdot \\ \cdot & \cdot & \cdot & 0 & 0 \\ \cdot & \cdot & \cdot & 0 & 0 \end{array}\right]\begin{bmatrix}\ddot{x}_G \\ \ddot{y}_G \\ \ddot{\theta} \\ R_{Ax} \\ R_{Ay}\end{bmatrix} = \begin{bmatrix}\text{known stuff} \\ \text{w/ } x_G, y_G, \\ \dot{x}_G, \dot{y}_G, \theta \\ \dot{\theta}, \text{parameters}\end{bmatrix}$$

(upper-left block: 3 × 3)

5×5

constraint eq